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(54) **MULTI-MODAL ENHANCEMENT OF LARGE LANGUAGE MODELS WITHOUT RETRAINING**

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(57) **ABSTRACT**

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A system and method for enhancing query responses from large language models without retraining by converting a query into a query vector; using a proximity metric to measure a proximity from the query vector to a plurality of vector embeddings stored in a vector database; ranking the plurality of vector embeddings based on proximity to the query vector; mapping the query to a homogenized context vector from a plurality of homogenized context vectors; using an augmented proximity metric to convert the proximity to an augmented proximity for each vector embeddings; performing an augmented ranking to refine the vector embeddings to those most relevant to the query; creating a prompt for a large language model comprising the query and the text data corresponding to refined vector embeddings as context; and feeding the prompt to the large language model to generate a response to the query.

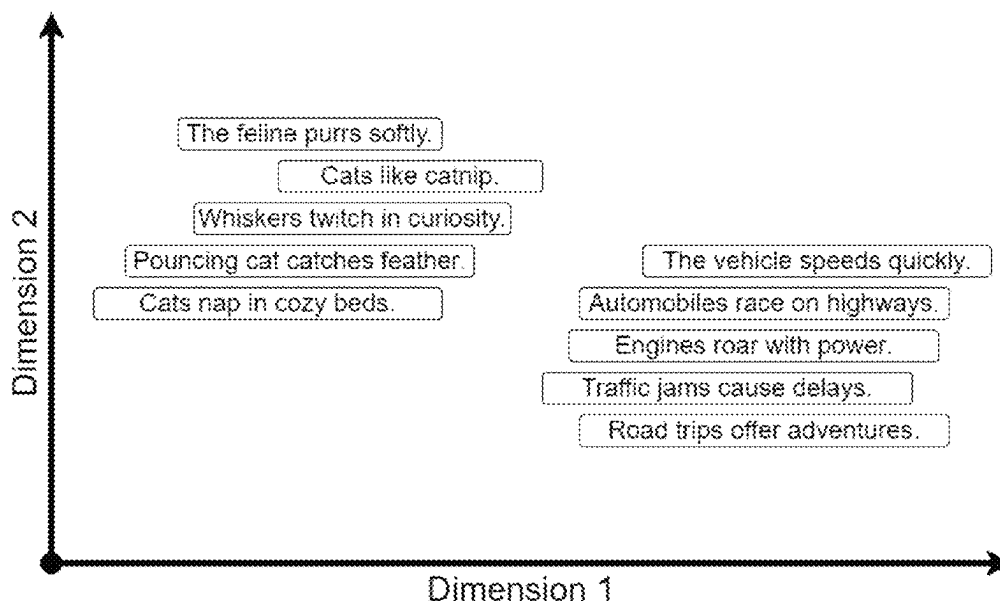
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See application file for complete search history.

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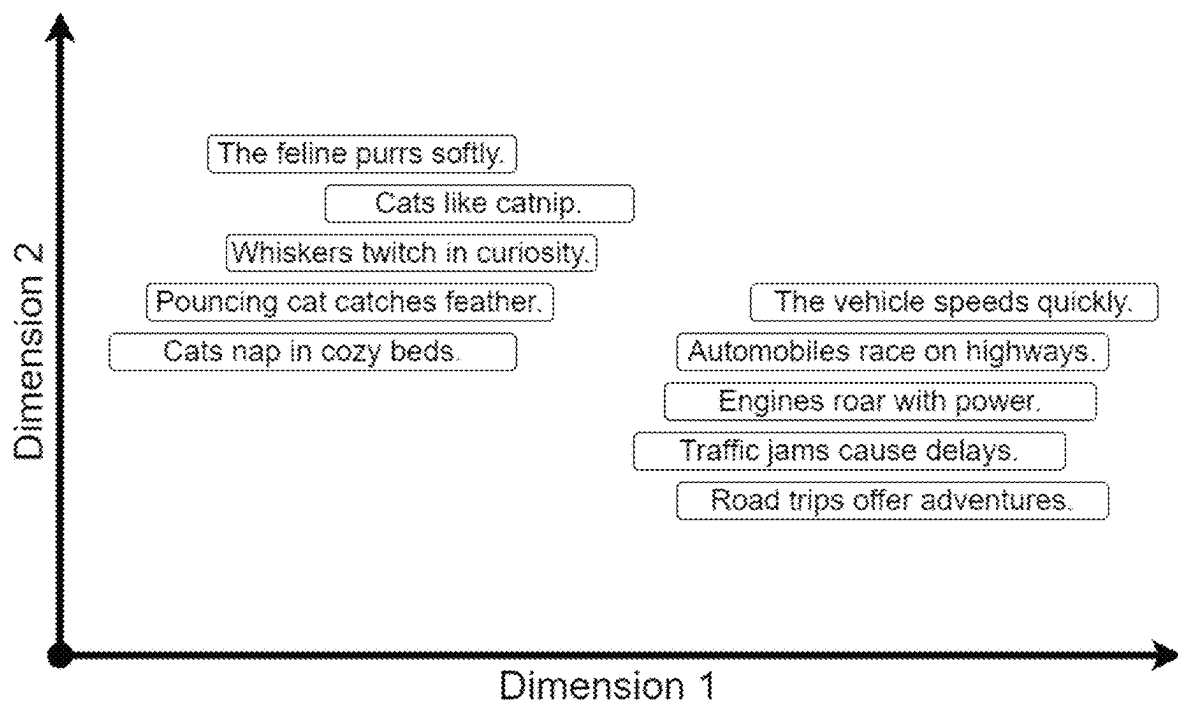


FIG. 1

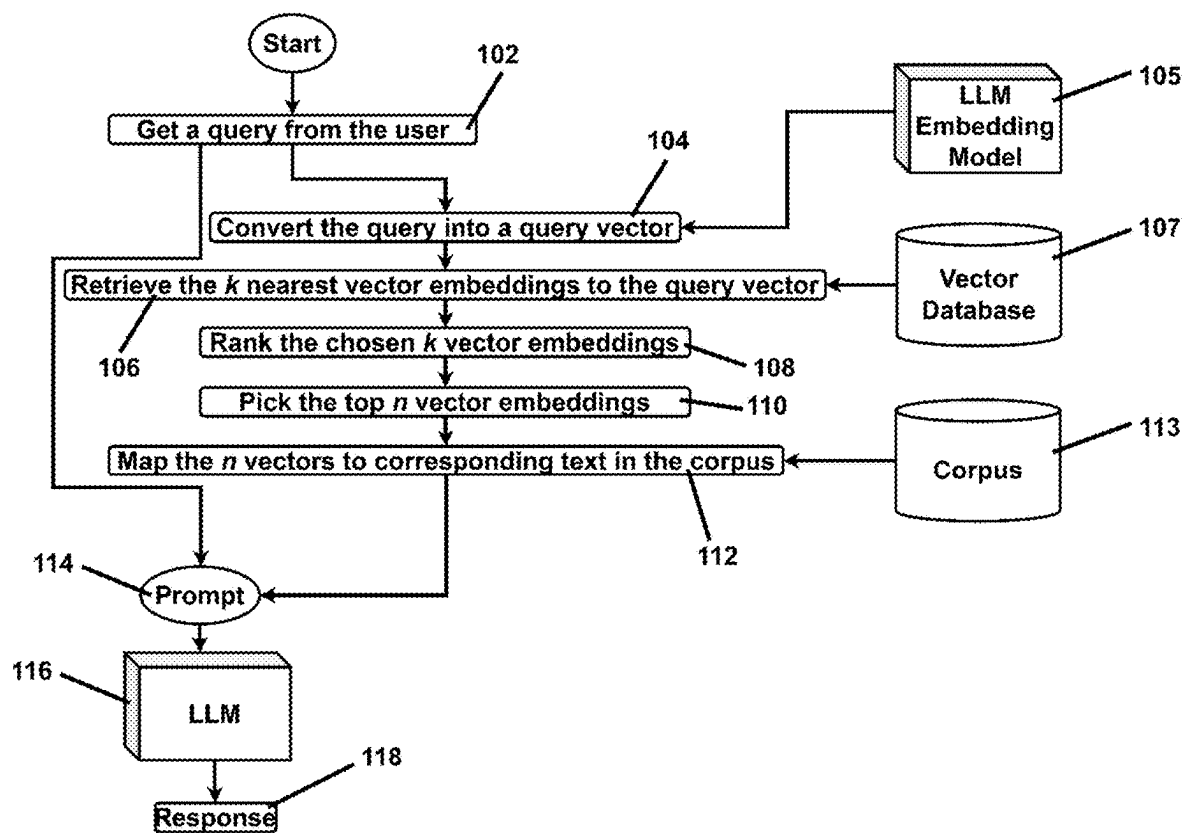


FIG. 2

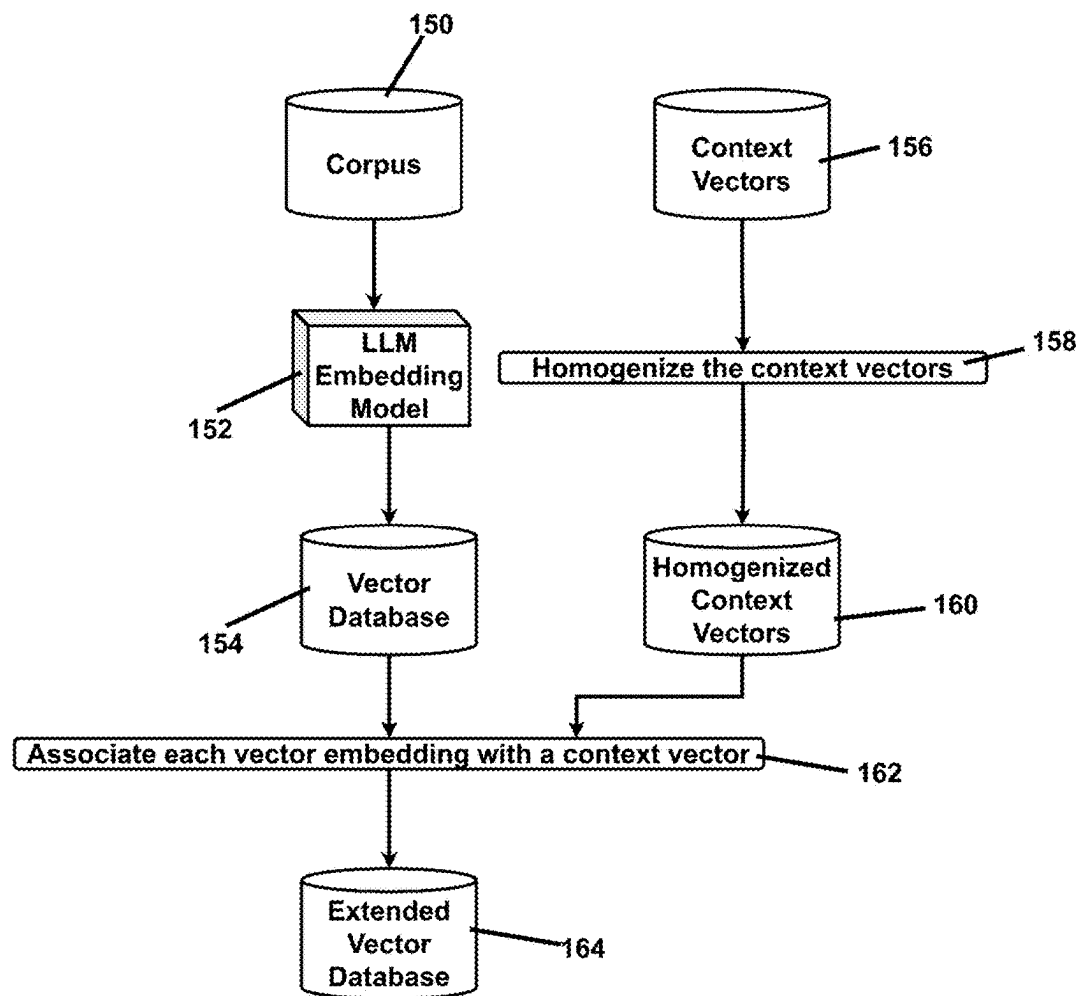


FIG. 3

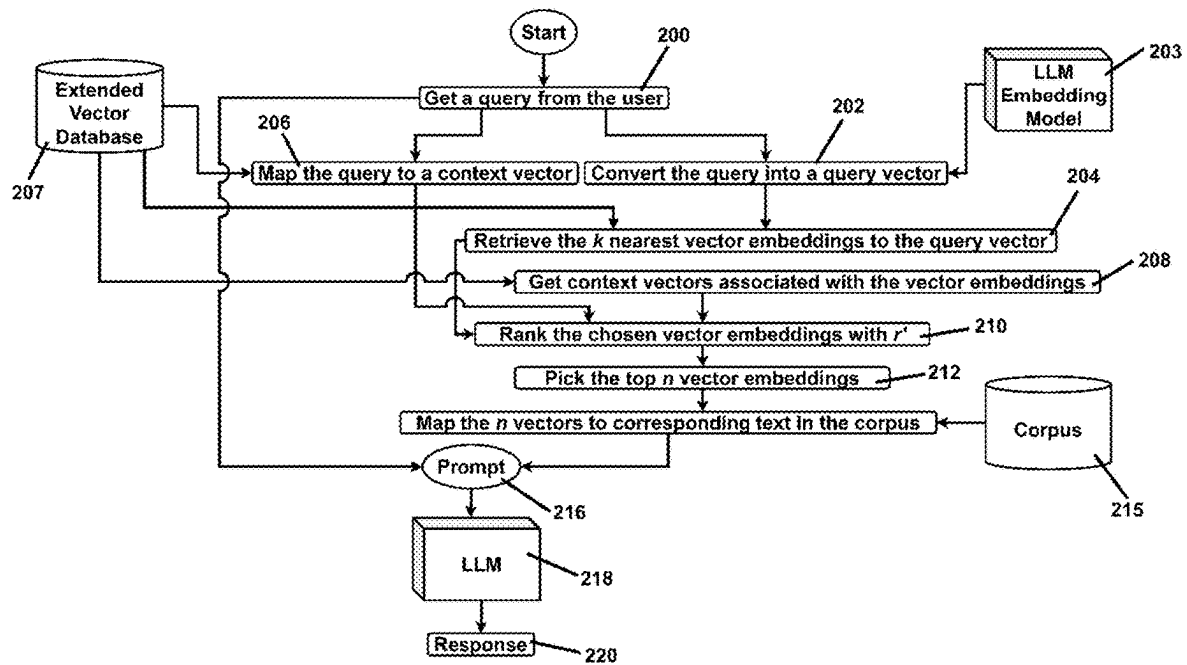
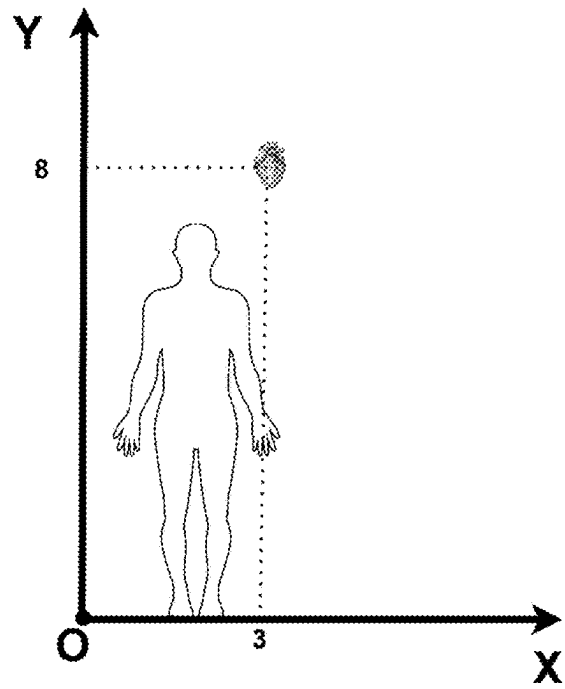
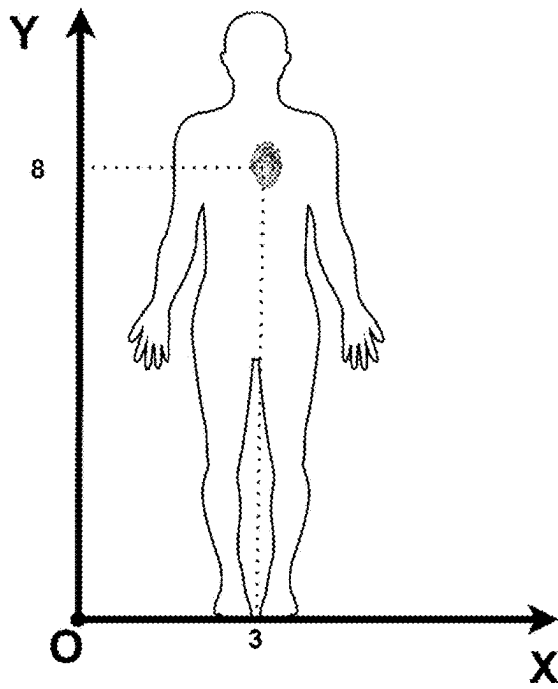


FIG. 4



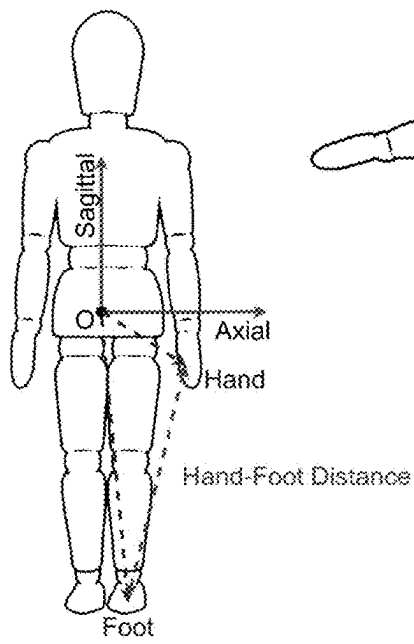


FIG. 6A

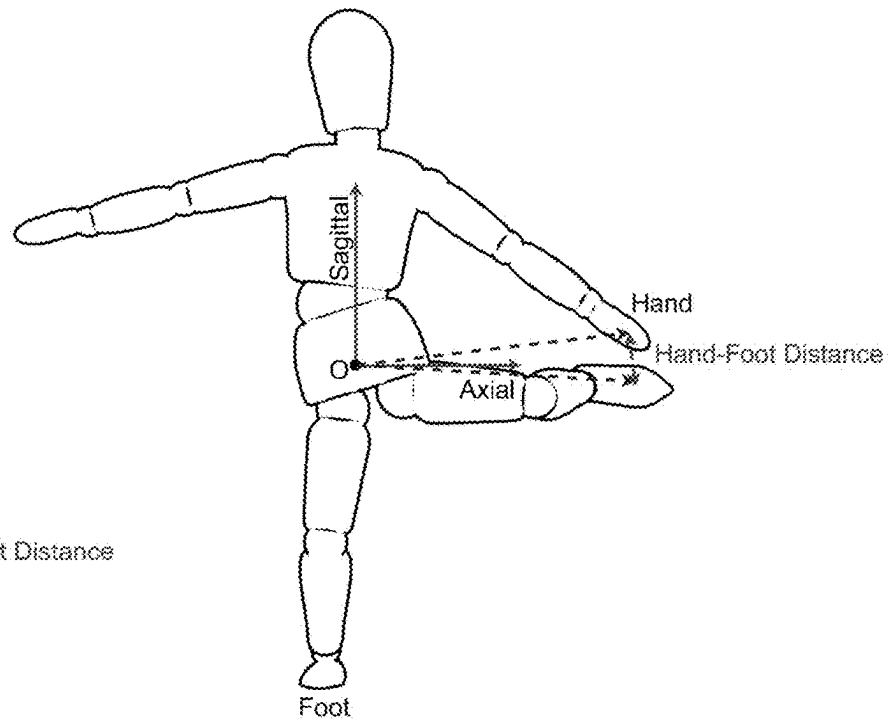


FIG. 6B

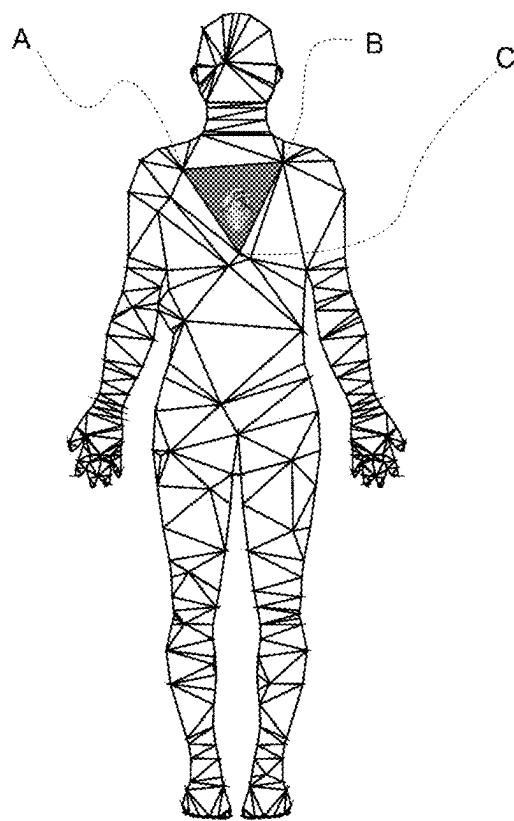


FIG. 7A

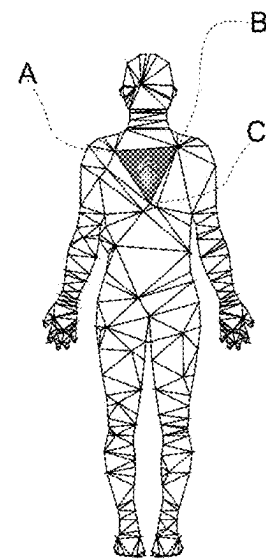
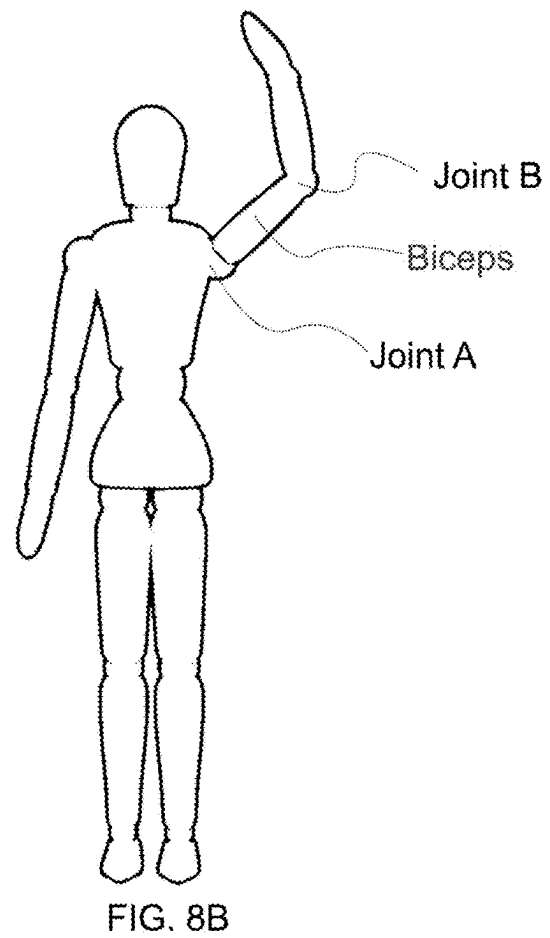
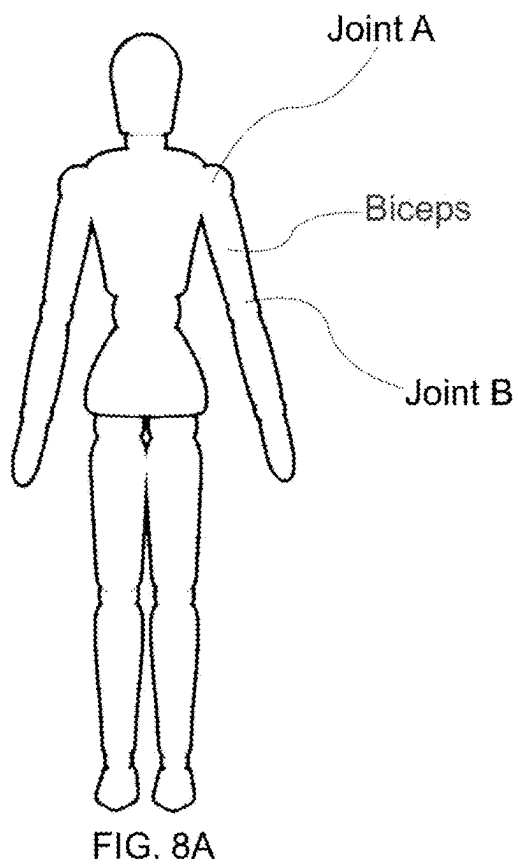


FIG. 7B



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MULTI-MODAL ENHANCEMENT OF LARGE LANGUAGE MODELS WITHOUT RETRAINING

CROSS REFERENCE TO RELATED APPLICATIONS

This patent application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/583,092, entitled “Enhancing Medical Query Responses from Large Language Models,” filed Sep. 15, 2023, which application is incorporated in its entirety here by this reference.

FIELD OF THE INVENTION

Generative AI applications that operate in a specific domain such as the medical field.

BACKGROUND

Recent Large Language Models (LLMs) have demonstrated remarkable capabilities in generating human-like answers to complex queries and providing information across a broad range of different domains of knowledge. The application of LLMs to generate relevant content is generally known as Generative Artificial Intelligence (Generative AI or GenAI for short). However, LLMs are also prone to delivering factually incorrect and sometimes purely fabricated replies. This phenomenon is often known as “hallucinations”. The problem of hallucinations is particularly acute for applications where misleading or unreliable answers can cause serious harm such as for medical decision-making. The challenge of producing more reliable and trustworthy LLMs is called alignment in the field of AI, and it is currently a source of major research and commercial development. One general solution practiced by vendors who develop foundational LLMs is to either improve the model architecture or the data used to train the model itself. This approach is simultaneously very resource-intensive, slow, and has shown relatively limited success in practice. Notably, achieving alignment through re-architecting and re-training the models provides no real guarantees that the final model will always behave as expected. Therefore, there is a need for better techniques to improve alignment in LLMs.

SUMMARY OF THE INVENTION

This invention presents a system and method that improves the alignment of LLMs: 1) without requiring changes to the architecture of the pre-trained LLMs, 2) without requiring extensive re-training of the model, and 3) allowing the integration of multi-modal information into existing models. One emerging approach that has proven successful in improving alignment is called Retrieval Augmented Generation (RAG) (Patrick S. H. Lewis et al., 2020) and it relies on two key ideas: 1) restricting the applicability of the LLM to a more specific domain of knowledge, and 2) leveraging vector search to ground the generative capabilities of the LLM to content that is known to be factually correct.

The notion of grounding refers to anchoring the text generated by the LLM to a specific set of content that is trusted. Vector search (also known as semantic search) exploits the notion that pre-trained LLMs can be used to

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niques with desirable statistical properties (see, FIG. 1). Vector embeddings that correspond to sentences that are conceptually or semantically related are closer to each other in this vector space than sentences that are conceptually or semantically unrelated. Vector embeddings of conceptually or semantically related concepts are well-clustered in the vector space. Clusters of conceptually related vector embeddings are well separated from clusters of unrelated vector embeddings as shown in FIG. 1.

In this regard, the notion of proximity or closeness of two vector embeddings is defined by a proximity metric. Mathematically, if we call the space of the vector embedding V , the proximity metric d is a function that takes two values in V and returns a scalar that measures how close two vectors are. Common choices of proximity metric are the dot product, cosine similarity, or the Euclidean distance. As such, the proximity metric can be a similarity metric or a distance metric. The difference between a similarity metric and a distance metric is the former needs to be maximized to improve proximity or closeness, while the latter needs to be minimized to improve proximity or closeness. However, the specific choice of proximity metric is not necessarily relevant to the scope of this invention and any choice that adequately quantifies the proximity of two vectors is valid for the present invention. In the remainder of this text, the measurement of closeness between two vectors is referred to as the proximity between the vectors with the understanding that a similarity metric, a distance metric, and the like, can be used for this purpose as long as the inequalities in the formulation are defined consistently.

Practitioners recently leveraged the concept of vector embeddings and vector search to implement a new generation of database systems called vector databases. In a vector database, text records are associated with the corresponding vector embedding and stored in persistent storage. When the user formulates a query in natural language, the vector database proceeds as follows: 1) convert the query text into a vector embedding (query vector) q , 2) use an algorithm, such as k -nearest neighbor search, to find k vectors in the database that are chosen as being the closest to the given query vector based on the chosen similarity metric, 3) present the text results corresponding to the k chosen vectors to the user.

Furthermore, the distance between the query vector and the k vectors found by the k -nearest neighbor search can be used directly to rank the results from most relevant to least relevant, the shorter the distance $d(q, v_i)$ being the more relevant and the longer distance $d(q, v_i)$ being the less relevant. Call $r: \Re \rightarrow \Re$ the ranking function, then we can associate a score s_i to each chosen vector v_i by computing the distance between q and v_i and using function r to obtain:

$$s_i \leftarrow r(d(q, v_i)).$$

Ranking the results amounts to sorting them based on the value of s_i , for example, from highest to lowest.

In some cases, practitioners may choose to employ a more advanced version of ranking functions sometimes known as neural reranking. With neural reranking: 1) the k nearest vector embeddings are chosen based on the proximity metric, 2) the neural reranker generates a score by comparing the textual representation of the query with the textual representation of the top k results. Neural rerankers are implemented by using machine learning techniques known to those skilled in the art where the model is trained on large number of text pairs to produce a score based on how closely related the pairs are.

Aside from delivering an entirely new and powerful type of database that can operate directly with natural language and match results based on meaning, the capabilities of vector databases have also proven very useful in improving the alignment of generative AI models through the aforementioned concept of Retrieval Augmented Generation (see, FIG. 2). To use RAG, developers first create a corpus of text data whose information is known to be trustworthy and store the data broken down by sentences (chunks) into a vector database. Then, when the user prompts the AI with a query, the system proceeds as follows: 1) receive a query from the user **102**, 2) convert the text query to a corresponding vector embedding **q 104** based on an LLM Embedding Model **105**, 3) retrieve the *k* nearest results to the given query **106** in the vector database **107**, 4) rank the results based on the relevance scores *s_i* from highest score to lowest score **108**, 5) pick the top *n* results **110**, 6) map the *n* vectors to corresponding text **112** in the corpus **113**, and 7) feed the top *n* results **114** as context for the LLM **116** accompanied by the original query **q 102** in the form of an appropriate natural language prompt, and 8) display the response **118**.

Assuming that the top *n* results $\{v_k, v_{k+1}, v_{k+2}, v_{k+n-1}\}$ are sufficiently consistent with the query *q*, the typical outcome of this process is that the LLM will ground the generated answer to the trusted knowledge of the corpus in the vector database. Without this kind of grounding, LLMs are more likely to generate results based on unreliable information that was present in the model's training data or on entirely fabricated notions. Steps 1-4 are called the retrieval phase and steps 5-7 are called the generation phase. Notably, LLMs, due to technical constraints, limit the number of tokens (smallest units of text used by LLMs) that can be used to formulate prompts, which in turn does not allow using an entire corpus of data as context. The benefit of RAG is that it overcomes this limitation in context size.

This approach works well as long as the retrieval phase truly produces results that are highly relevant to the query *q*. The latter is not guaranteed. If the query is not related to the domain of knowledge encompassed by the corpus, then the *k* nearest results are simply the least unrelated ones, but they may not be conceptually consistent with the query posed by the user. If the LLM has not received sufficient training in the specific domain encompassed by the corpus (e.g. medical literature), then the quality of the vector embeddings related to the chosen domain may be low and the distance between vector embeddings may not always correlate well to concepts that are semantically close to each other. The primary limitations of this approach are that: 1) the ranking function may not boost the most relevant results to lie within the top *n* results, and 2) the top *n* results may contain unexpected outliers that are not conceptually related to the given query.

If these undesirable conditions occur, then even with RAG the LLM may produce inconsistent or unexpected results (in most cases the text is still grounded to the trusted corpus, so the generated text will be generally factually correct, but may not be adequately related to the query). The purpose of this invention is to augment the vector embeddings, the proximity metric, and the ranking function to substantially alleviate the occurrence of the shortcomings mentioned above and thus improve the alignment of generative AI without requiring re-training of the underlying LLM.

The invention comprises associating each vector embedding with another modality of information (e.g., positional data, geolocation, color space data, functional data, financial profiles, and the like) according to conditions and con-

straints defined in detail later in the document. The elements of this additional modality of information is referred to herein as context vectors. If the new context vectors are expressed in such a way that one can associate a given query with the appropriate context vector, then one can augment the original vector embeddings with the context vectors, appropriately augment the proximity metric, and augment the ranking function to obtain a new extended form of RAG that leverages more accurate contextual information when generating results.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows an illustration of the clustering of vector embeddings for short sentences related to the concepts of "cat" and "car" in two hypothetical embedding dimensions (real vector embeddings typically have over a thousand dimensions). The training of the LLM moves sentences that are conceptually related close to each other in the vector space but keeps unrelated concepts well-separated.

FIG. 2 is a flow diagram of a Retrieval Augmented Generation.

FIG. 3 is a flow diagram illustrating how to create the extended vector database.

FIG. 4 is a flow diagram of an embodiment of the present invention.

FIGS. 5A-5B are illustrations representing the position of the heart in the body using a world coordinate frame for two different body sizes. The location of the heart with respect to the body is not invariant to changes in body habitus H.

FIGS. 6A-6B are illustrations showing the position of the hand and foot represented with respect to a body-aligned reference frame centered at the pelvis. The distance between the hand and foot is not invariant to changes in the pose J.

FIGS. 7A-7B are 2D illustrations showing the location of the heart expressed with respect to a local domain (the area inside triangle ABC). In the simplified 2D case, the simplex is the triangle. Note that the position of the heart is expressed consistently in terms of the vertices A,B,C and it is invariant under changes in body habitus.

FIGS. 8A-8B are illustrations of the location of the biceps muscle expressed as the midpoint between two joint landmarks. Clearly the location of the biceps muscle expressed this way is invariant under changes in pose J.

DETAILED DESCRIPTION OF INVENTION

The detailed description set forth below in connection with the appended drawings is intended as a description of presently-preferred embodiments of the invention and is not intended to represent the only forms in which the present invention may be constructed or utilized. The description sets forth the functions and the sequence of steps for constructing and operating the invention in connection with the illustrated embodiments. It is to be understood, however, that the same or equivalent functions and sequences may be accomplished by different embodiments that are also intended to be encompassed within the spirit and scope of the invention.

The invention of the present application is a system and method for improving responses to queries to large language models (LLM) without having to retrain the LLM. As a preferred embodiment, the invention is described in the context of medical knowledge, but the presented system and method can be applied to many other domains of knowledge without departing from the spirit of the invention, such as law, engineering, math, science, philosophy, sociology, his-

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tory, literature, and the like. We start from a corpus C of trusted textual data, $c_i \in C$, curated by known experts. For example, experts in the field of medicine, law, engineering, math, science, philosophy, sociology, history, literature, and the like, would create the textual data of facts, concepts, ideas, opinions, theories, laws, hypotheses, proposals, and the like. The textual data c_i can be converted into vector embeddings $v_i \in V$ using an embedding model E , where $v_i = E(c_i)$. As such, a plurality of vector embeddings can be stored in a vector database. Similarly, a query from a user can be converted into a query vector q .

One can compute the proximity between two vector embeddings in V , and between a vector embedding in V and the query vector, with a properly chosen vector proximity metric d_v . The proximity metric d_v is a measure of the closeness of units of data being compared. As such, the vector proximity metric d_v can be a measure of numerical distance, geometric distance, semantic similarity, vector alignment, conceptual similarity, visual similarity, auditory similarity, and the like. The proximity between two vector embeddings or between a query vector and a vector embedding can be considered a raw measurement. As such, if the proximity measure is distance, then the proximity metric would measure a raw distance between two vector embeddings or between a vector embedding and the query vector.

In the present invention, an additional modality of information is selected that may or may not be textual, but is strongly correlated to the domain of knowledge of the corpus. This is called the extended context, P . The elements of P , $p_j \in P$ are the context vectors p_j and the set P , which is endowed with an augmented proximity metric d_p (a metric that is appropriately defined for measuring the closeness of context vectors). Based on one's domain knowledge, the practitioner or an appropriately defined algorithm associates the query vector and/or each vector embedding $v_i \in V$ with a conceptually related context vector $p_j \in P$. For example, let M be the function that maps v_i to p_j using known techniques: $p_j = M(v_i)$.

In general, the set P is chosen so that: $|P| \leq |V| = |C|$ (Eq. 1).

In one embodiment, the problem of mapping queries to context vectors p_j can be formulated as a problem of classification. Hence, to make the classification scalable and manageable by an LLM, the size of the set P needs to be kept small, and specifically, smaller than the set of vector embeddings V . In other words, a given value p_j may be associated with multiple values of v_i , which in turn are in a one-to-one mapping with text data c_i . Given a query vector q , and set of k vectors $v_i^q \in V^q$ (k can be any predetermined number) that are found to be closest to q by the proximity metric d_v (these are the candidates vector embeddings that could potentially be used in the prompt), the purpose of the context vectors p_j is to boost (i.e. increase) or bury (i.e. decrease) the rankings of the tuple (q, v_i^q) to improve the quality of the results produced by vector search and identify the most relevant vector embeddings to use in the prompt.

By way of example only, in the medical field, the corpus C can contain a collection of text paragraphs describing pathologies of each organ in the human body. Each of these text paragraphs would be associated with vector embeddings and make up the total vector embeddings in the vector database. For instance, paragraphs c_a , c_b may describe pathologies affecting the tonsils, and paragraph c_c may describe possible ways the appendix can get inflamed. Given the query "What are common causes of tonsillitis?" and its corresponding vector embedding q , we expect the vector database to rank vectors v_a, v_b that correspond to paragraphs c_a, c_b higher (i.e. closer) than other vectors that do not relate

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to the tonsils, such as vector v_c . However, it may occur in practice that the proximity $d_v(q, v_c)$ is higher than both proximities $d_v(q, v_a)$ and $d_v(q, v_b)$ so that the top result of vector search produced by applying the ranking function to the aforementioned distances will be v_c , which would not be desirable. In other words, if the set of k vectors was predetermined to be 3, and $d_v(q, v_c)$, $d_v(q, v_a)$ and $d_v(q, v_b)$ were the top three proximal vectors to the query vector, then v_c , v_a , and v_b would be the set of candidate vector embeddings even though v_c does not appear to be relevant. Given the inherent complexities and opaque nature of LLMs it is generally not possible to troubleshoot the root cause of such unexpected results. Perhaps the embedding may place tonsils and the appendix close to each other in the embedding space because both are likely to get inflamed and they may have appeared in similar contexts in the training data of the model.

Since users cannot easily alter the LLM that produces the embeddings to facilitate the expected outcome, the present invention utilizes the context vectors p_j to reliably alter the rankings and boost results related to tonsils to queries asking about tonsils. By way of example only, we can define the context vectors p_j as homogenized coordinates (see section on "Coordinate Homogenization") of the location of each organ in the body and associate the vector embeddings of each paragraph with the corresponding location on the body, such that:

$$v_a = E(c_a) \rightarrow p_1 = M(v_a)(\text{location of tonsils})$$

$$v_b = E(c_b) \rightarrow p_1 = M(v_b)(\text{location of tonsils})$$

$$v_c = E(c_c) \rightarrow p_2 = M(v_c)(\text{location of appendix})$$

Therefore, each vector embedding v is now associated with a context vector p . The query q can also be mapped to the same vector space of P using function M (see section on "Mapping Function" for more details): $q^p = M(q)$, which for the query "What are common causes of tonsillitis?" is the location of tonsils: $q^p = p_1$.

Assuming that the proximity metric is chosen to be a distance metric where the distance of a vector with itself is zero, $d(v, v) = 0$; then prior to applying the context vector, the raw distance would be as follows:

$$d_v(q, v_c) < d_v(q, v_a)$$

$$d_v(q, v_c) < d_v(q, v_b)$$

which is the undesirable condition that leads appendix to be chosen for a query about the tonsils.

But, by using the augmented proximity metric, by virtue of introducing the context vectors and a mapping function, we obtain that:

$$d_p(M(q), M(v_c)) = d_p(q^p, p_2) = d_p(p_1, p_2) > 0$$

$$d_p(M(q), M(v_a)) = d_p(q^p, p_1) = d_p(p_1, p_1) = 0$$

$$d_p(M(q), M(v_b)) = d_p(q^p, p_1) = d_p(p_1, p_1) = 0$$

Hence, the augmented distances are:

$$d_P(M(q), M(v_c)) > d_P(M(q), M(v_a))$$

$$d_P(M(q), M(v_c)) > d_P(M(q), M(v_b))$$

Therefore, according to the augmented proximity metric d_P , paragraphs related to tonsils c_a, c_b are closer to the given query than the unrelated paragraph c_c discussing the inflammation of the appendix.

At this point, a new augmented ranking function $r': \mathcal{R} \times \mathcal{R} \rightarrow \mathcal{R}$ is defined that uses a blend of both distance metrics d_V and d_P to score the results from vector search in a way that is more consistent with our expectations for the given domain of knowledge to generate a more relevant set of vector embeddings from the candidate of vector embeddings:

$$s'_i \leftarrow r'(d_V(q, v_i), d_P(q, M(v_i)))$$

(an adequate definition for r' is discussed later in this section). The core advantage of this approach is that for many domains of knowledge there exist many effective choices of extended context (e.g., location of body parts and organs for medical text) so that distances between its elements are always strongly correlated to semantic closeness. This contrasts with proximity measured in the space of vector embeddings where their correlation to semantic closeness is defined by the LLM embedding model probabilistically and it is not entirely predictable. In the example above, applying a context vector to the query vector and the 3 candidate vectors would result in a refinement of the ranking such that the vector embedding associated with tonsils (v_c) would be much further from the query vector or buried deeper in the ranking so as to fall out of the set of candidate vector embeddings, resulting in a set of relevant vector embeddings that one would expect.

More formally, given a space of vector embeddings V , in order to properly define the extended context, a vector space is chosen that is relevant to the domain of expertise in a desired field (e.g., medical field) and satisfies the following properties: 1) A space of vectors $p_j \in P$, 2) A measure of closeness d_P , 3) An augmented ranking function r' , and 4) A mapping function $M: V \rightarrow P$, such that for every query vector q :

$$d_P(M(q), p_i) < d_P(M(q), p_j) \Rightarrow r'(d_V(q, v_i), d_P(M(q), p_i)) > r'(d_V(q, v_j), d_P(M(q), p_j)) \quad (\text{Eq. 2})$$

where $p_i = M(v_i)$ and $p_j = M(v_j)$.

This last condition stipulates that if a context vector p in P is closer to a given query q , then the augmented ranking function r' should give a higher score to the vector embedding corresponding to the context vector p . A general choice of r' that may be appropriate for most applications is a linear blend of the proximity metrics for vector embeddings and context vectors:

$$r'(d_V(q, v), d_P(M(q), p)) = -((1-\alpha)d_V(q, v) + \alpha d_P(M(q), p))$$

where α is a tuning parameter chosen appropriately to satisfy the condition in equation 2 above. Parameter α may be a fixed value chosen manually or it may be dynamically chosen by a machine learning algorithm trained specifically to improve the quality of the ranking for a broad range of data points to identify a more relevant set of vector embeddings. If the proximity metric is chosen to be a similarity metric, then the sign of r' would be flipped.

In another embodiment, each context vector is associated with a confidence bound. The confidence bound is used to weigh the proximity measure. For instance, if the context vector is a small organ like the gallbladder which is well localized in a particular region of the body, its influence should be higher in the computation of the proximity metric, than a larger organ like the skin, which is distributed across the body and not localized in a specific region.

In summary, the invention comprises a system and method made of two parts: First, create an extended vector database as shown in FIG. 3) as follows: Given a corpus of knowledge **C 150** defined by a collection of text, compute a vector embedding **152** using known techniques for each item in the collection via the LLM embedding model to create a vector database **154**. Introduce another modality of information (extended context) **156** appropriate to the domain of knowledge defined by the corpus **C** (this choice must be made the practitioner skilled in the art and knowledgeable of the domain) where: i) values can be readily defined numerically as vectors (context vectors), and ii) distances between vectors correlate well with semantic similarity. Homogenize the context vectors **158** so that they define a consistent measure of closeness if needed (this is explained in the Section "Coordinate Homogenization") to create a homogenized context vector database **160**. Map each homogenized context vectors with one or more vector embedding **162** (see section on "Mapping Function") to create the extended vector database **164**.

Second, perform an extended version of Retrieval Augmented Generation (see, FIG. 4) as follows: Given a query **200**, compute the corresponding query vector **202** using the embedding function (for example using the embedding model associated with an LLM **203**). Find the k nearest vector embeddings to the query vector **204**. From the same query, determine the corresponding context vector **206** using function M (see section on "Mapping Function") based on the extended vector database **207**. Retrieve the context vectors associated with the k vector embeddings **208**. Apply the augmented ranking function to the information gathered thus far to assign scores to each of the k results **210**. Sort the results based on the ranking score (e.g., from largest to smallest) and select a predetermined number of the top vector embeddings **212**. Map the top vector embeddings to the corresponding text in the corpus **214** from the corpus of text **215**. Create a prompt **216** for a large language model comprising the query and the text data corresponding to the set of relevant vector embeddings as context. Feed the prompt to the large language model **218** to generate a response **220** to the query. Display the response to the user.

In light of the aforementioned description, a method for reducing hallucinations or improving relevance of a response to a query in a large language model can comprise the steps of: receiving a query from a user; converting the query into a query vector; in a vector database comprising a plurality of vector embeddings, using a proximity metric to measure a raw distance from the query vector to each vector embedding in the plurality of vector embeddings, wherein each vector embedding in the plurality of vector embeddings corresponds with a text data; comparing the raw distance from the query vector to the plurality of vector embeddings; evaluating a ranking function to order the plurality of vector embeddings to the query vector based on distance; retrieving a predetermined number of vector embeddings to establish a set of candidate vector embeddings that are most proximal to the query vector (e.g., their similarity measure is higher or their distance measure is lower) compared to the remaining vector embeddings in the plurality of vector embeddings.

dings; mapping the query to one homogenized context vector from a plurality of homogenized context vectors; using an augmented proximity metric to convert the raw proximity for each of the candidate vector embeddings to an augmented proximity for each of the candidate vector embeddings; performing an augmented ranking to refine the set of candidate vector embeddings to a set of relevant vector embeddings; creating a prompt for a large language model comprising the query and the text data corresponding to the set of relevant vector embeddings as context; and feeding the prompt to the large language model to generate a response to the query.

The step of using a proximity metric to measure a raw distance from the query vector to each vector embedding and selecting the best vector embeddings can comprise maximizing the proximity metric. Maximizing the proximity metric comprises maximizing a similarity metric or minimizing a distance metric.

The step of mapping the query to the homogenized context vector can comprise using the large language model to classify the query based on a text label associated with the homogenized context vector. Each homogenized context vector is obtained by re-scaling a raw context vector so that the augmented proximity vector is defined consistently irrespective of a parameterization of a domain as described in further detail below regarding coordinate homogenization. The raw context vector can be homogenized using local domains, such as bounded geometric shapes that partition the overall space. In some embodiments, the raw context vector can be homogenized using landmarks. For example, in the medical field, the landmarks can be anatomical body parts.

Preferably, before using the augmented proximity metric, each vector embedding is associated with one of the homogenized context vectors of the plurality of homogenized vectors and stored in an extended vector database.

Mapping Function

This section explains how to define the function M for mapping vector embeddings to context vectors. Given that the sets V and P are discrete and finite, the association between vector embeddings v_i and context vectors p_j can be achieved through a simple lookup table. However the process of mapping a query vector to a context vector in P is not trivial. One may try to frame it as an optimization problem; if the proximity measure is a similarity measure, given a finite set of context vectors $\{p_1, p_2, \dots, p_n\}$ find the value q^P such that:

$$q^P = \operatorname{argmax}_{p_j} d_r(q, M^{-1}(p_j)),$$

where M^{-1} is the inverse mapping of M . Alternatively, if the proximity measure is distance measure, find the value q^P such that

$$q^P = \operatorname{argmax}_{p_j} d_r(q, M^{-1}(p_j)).$$

This equation, however, while being well-defined, does not have a unique solution in general. As stated in equation 1 multiple values of V may map to the same value of P , which means that $M^{-1}(p_j)$ is not unique.

Assuming that the cardinality of P , $|P|$ is small, the problem of mapping q to q^P can be alternatively cast as a

problem of classification. Instead of considering the query in the space of the vector embeddings,

$$q = E(\bar{q})$$

we operate on the original textual version of the query $\bar{q}$ directly. Furthermore, let $\{\bar{p}_1, \bar{p}_2, \dots, \bar{p}_n\}$ be a text label associated with the context vectors $\{p_1, p_2, \dots, p_n\}$. Knowing that LLMs are particularly good at classifying text within a finite number of classes, the mapping between q and q^P can be found directly by using a prompt of the form: Given the query “ q ” find the class that best describes this query among the following classes: $\{\bar{p}_1, \bar{p}_2, \dots, \bar{p}_n\}$. Answer with a single word that indicates the chosen class or “UNKNOWN” if none of the classes is adequate.

The output of this prompt from the LLM will give the text label for the context of the query from which q^P can easily be derived.

Coordinate Homogenization

For the method described above to work correctly it is essential that the choice of context vectors $p_j \in P$ is such that the measure of proximity d_P can be defined consistently over the vector space. In many cases, such as in the case of geolocation data, this is not a problem. However, when the context vectors are defined on a field that can vary from one data acquisition of the context vectors to another, the metric may not be consistent and the raw distance between the same two conceptually related context vectors may vary substantially over the data set.

One notable example of this problem is radiological data. There is no well-defined coordinate system that allows defining the position and orientation of body organs consistently. Common choices of coordinate frames result in coordinates varying upon changes in the pose and/or body habitus of the patient. This section describes a solution to this problem. FIGS. 5 and 6 illustrate why a world coordinate frame or a body-aligned coordinate frame will not be satisfactory for this application.

Let $B(J, H)$ be a function that yields the geometry of a human body given a parameterization of the pose J and the body habitus H , where B is the domain (body) and J and H are the parameterizations. What is sought is a homogenization or mapping of the context vectors $p_j \in P$, such that the distance between any two context vectors $d_P(p_i, p_j)$ is the same for all values of J and H . In other words, the context vectors are homogenized context vectors to account for different body sizes or different poses of a body. One approach is to discretize the body into a set of local domains defined as simplices (see, FIG. 7) and express the location of each body organ as a vector of the form:

$$(m, x, y, z, r_x, r_y, r_z)$$

where:

m is an index identifying the local domain

x, y, z are barycentric coordinates that express the location of the organ with respect to the local domain m

r_x, r_y, r_z are a parametrization of the orientation of the organ with respect to a reference edge of the local domain.

One could define an equivalent vectorization of the location of a body organ using any other valid parameterization of position, such as trilinear coordinates, or rotations, such as quaternions. For example:

$$(m, x, y, z, q_x, q_y, q_z, q_w)$$

where q_x, q_y, q_z, q_w are the components of a normalized quaternion.

The local domains can be generated using techniques known in the art such as the Delaunay tetrahedralization algorithm. One challenge with this approach is that there is no straightforward way of ensuring that the partitioning into local domains of the body stays consistent as the parameters J and H change.

An alternate approach is to use well-established anatomical landmarks as reference points (e.g. the location of joints in the human skeleton) as reference positions that stay consistent when varying the pose J and habitus H of the body (see, FIG. 8). Given a choice of anatomical landmarks then one can define the position of organs as linear combinations of the landmark positions and the rotations as changes in orientation with respect to the coronal, sagittal, and axial planes of the body.

System Architecture

The system can take the form of an entirely hardware embodiment, an entirely software embodiment or an embodiment containing both hardware and software elements. As such, the system of the present invention can comprise a server connected to a network, the server receiving product requests from users via the network, the server comprising at least one processor; a database for storing product information; and a memory operatively coupled to the processor, the memory storing program instructions that when executed by the processor, causes the processor to perform one or more of the functions and features described herein. In one embodiment, the system is implemented in software, which includes but is not limited to firmware, resident software, microcode, etc., configured to perform one or more of the functions and features described herein.

Furthermore, the system can take the form of a computer program product accessible from a computer-usable or computer-readable medium providing program code for use by or in connection with a computer or any instruction execution system configured to perform any one or more of the functions and features described herein. For the purposes of this description, a computer-usable or computer readable medium can be any apparatus that can contain, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device.

The medium can be an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system (or apparatus or device) or a propagation medium. Examples of a computer-readable medium comprise a semiconductor or solid-state memory, magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk and an optical disk. Current examples of optical disks comprise compact disk-read only memory (CD-ROM), compact disk-read/write (CD-R/W) and DVD.

A data processing system suitable for storing and/or executing program code comprises at least one processor coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories that provide temporary storage of at least some program code in order to reduce the number of times code is retrieved from bulk storage during execution.

Input/output or I/O devices (including but not limited to keyboards, displays, pointing devices, etc.) can be coupled to the system either directly or through intervening I/O controllers.

Network adapters may also be coupled to the system to enable the data processing system to become coupled to

other data processing systems or remote printers or storage devices through intervening private or public networks. Modems, cable modem and Ethernet cards are just a few of the currently available types of network adapters.

Described above, aspects of the present application are embodied in a World Wide Web ("WWW") or ("Web") site accessible via the Internet. As is well known to those skilled in the art, the term "Internet" refers to the collection of networks and routers that use the Transmission Control Protocol/Internet Protocol ("TCP/IP") to communicate with one another. The internet can include a plurality of local area networks ("LANs") and a wide area network ("WAN") that are interconnected by routers. The routers are special purpose computers used to interface one LAN or WAN to another. Communication links within the LANs may be wireless, twisted wire pair, coaxial cable, or optical fiber, while communication links between networks may utilize 56 Kbps analog telephone lines, 1 Mbps digital T-1 lines, 45 Mbps T-3 lines or other communications links known to those skilled in the art.

Furthermore, computers and other related electronic devices can be remotely connected to either the LANs or the WAN via a digital communications device, modem and temporary telephone, or a wireless link. It will be appreciated that the internet comprises a vast number of such interconnected networks, computers, and routers.

As the Internet has grown, so has the WWW. As is appreciated by those skilled in the art, the WWW is a vast collection of interconnected or "hypertext" documents written in HTML, or other markup languages, that are electronically stored at or dynamically generated by "WWW sites" or "Web sites" throughout the Internet. Additionally, client-side software programs that communicate over the Web using the TCP/IP protocol are part of the WWW, such as JAVA® applets, instant messaging, e-mail, browser plug-ins, Macromedia Flash, chat and others. Other interactive hypertext environments may include proprietary environments such as those provided in America Online or other online service providers, as well as the "wireless Web" provided by various wireless networking providers, especially those in the cellular phone industry. It will be appreciated that the present application could apply in any such interactive communication environments, however, for purposes of discussion, the Web is used as an exemplary interactive hypertext environment with regard to the present application.

A website is a server/computer connected to the Internet that has massive storage capabilities for storing hypertext documents and that runs administrative software for handling requests for those stored hypertext documents as well as dynamically generating hypertext documents. Embedded within a hypertext document are a number of hyperlinks, i.e., highlighted portions of text which link the document to another hypertext document possibly stored at a website elsewhere on the Internet. Each hyperlink is assigned a URL that provides the name of the linked document on a server connected to the Internet. Thus, whenever a hypertext document is retrieved from any web server, the document is considered retrieved from the World Wide Web. Known to those skilled in the art, a web server may also include facilities for storing and transmitting application programs. Likewise, a web server may also include facilities for executing scripts and other application programs on the web server itself.

A remote access user may retrieve hypertext documents from the World Wide Web via a web browser program. A web browser is a software application program for providing a user interface to the WWW. Upon request from the remote

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access user via the web browser, the web browser requests the desired hypertext document from the appropriate web server using the URL for the document and the hypertext transport protocol ("HTTP"). HTTP is a higher-level protocol than TCP/IP and is designed specifically for the requirements of the WWW. HTTP runs on top of TCP/IP to transfer hypertext documents and user-supplied form data between server and client computers. The WWW browser may also retrieve programs from the web server, such as JAVA applets, for execution on the client computer. Finally, the WWW browser may include optional software components, called plug-ins, that run specialized functionality within the browser.

The foregoing description of the preferred embodiment of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the invention not be limited by this detailed description, but by the claims and the equivalents to the claims appended hereto.

The invention claimed is:

1. A method for enhancing query responses from large language models without retraining, the method comprising the steps of:

- a) receiving a query from a user;
- b) converting the query into a query vector;
- c) in a vector database comprising a plurality of vector embeddings, using a proximity metric to measure a raw distance from the query vector to each vector embedding in the plurality of vector embeddings, wherein each vector embedding in the plurality of vector embeddings corresponds with a text data;
- d) comparing the raw distance from the query vector to the plurality of vector embeddings;
- e) evaluating a ranking function to order the plurality of vector embeddings to the query vector based on distance;
- f) retrieving a predetermined number of vector embeddings to establish a set of candidate vector embeddings that are most proximal to the query vector compared to the remaining vector embeddings in the plurality of vector embeddings;
- g) mapping the query to one homogenized context vector from a plurality of homogenized context vectors;
- h) using an augmented proximity metric to convert the raw proximity for each of the candidate vector embeddings to the query vector to an augmented proximity for each of the candidate vector embeddings and their associated homogenized context vectors to the query vector and its mapped homogenized context vector, wherein each homogenized context vector is obtained by re-scaling a raw context vector so that an augmented proximity vector is defined consistently irrespective of a parameterization of a domain;
- i) performing an augmented ranking to refine the set of candidate vector embeddings to a set of relevant vector embeddings;
- j) creating a prompt for a large language model comprising the query and the text data corresponding to the set of relevant vector embeddings as context; and
- k) feeding the prompt to the large language model to generate a response to the query.

2. The method of claim 1, wherein the step of mapping the query to one homogenized context vector, comprises maximizing the proximity metric.

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3. The method of claim 2, wherein maximizing the proximity metric comprises maximizing a similarity metric.

4. The method of claim 2, wherein maximizing the proximity metric comprises minimizing a distance metric.

5. The method of claim 1, wherein the step of mapping the query to the homogenized context vector, comprises using the large language model to classify the query based on a text label associated with the homogenized context vector.

6. The method of claim 1, wherein the raw context vector is homogenized using local domains.

7. The method of claim 1, wherein the raw context vector is homogenized using landmarks.

8. The method of claim 1, wherein before using the augmented proximity metric, each vector embedding is associated with one of the homogenized context vectors of the plurality of homogenized vectors and stored in an extended vector database.

9. A system for enhancing query responses from large language models without retraining, comprising:

a server connected to a network, the server receiving product requests from users via the network, the server comprising: at least one processor; a database for storing information; and a memory operatively coupled to the processor, the memory storing program instructions that when executed by the processor, causes the processor to:

- a) receive a query from a user;
- b) convert the query into a query vector;
- c) use a proximity metric to measure a raw distance from the query vector to a plurality of vector embeddings stored in a vector database, wherein each vector embedding in the plurality of vector embeddings corresponds with a text data;
- d) compare the raw distance from the query vector to the plurality of vector embeddings;
- e) evaluate a ranking function to order the plurality of vector embeddings to the query vector based on proximity;
- f) retrieve a predetermined number of vector embeddings to establish a set of candidate vector embeddings that are most proximal to the query vector compared to the remaining vector embeddings;
- g) map the query to one homogenized context vector from a plurality of homogenized context vectors;
- h) use an augmented proximity metric to convert the raw distance to an augmented distance for each of the candidate vector embeddings, wherein the homogenized context vector is obtained by re-scaling a raw context vector so that a proximity vector is defined consistently irrespective of a parameterization of a domain;
- i) perform an augmented ranking to refine the set of candidate vector embeddings to a set of relevant vector embeddings;
- j) create a prompt for a large language model comprising the query and the text data corresponding to the set of relevant vector embeddings as context; and
- k) feed the prompt to the large language model to generate a response to the query.

10. The system of claim 9, wherein the mapping the query to the context vector, comprises maximizing the proximity metric.

11. The system of claim 10, wherein maximizing the proximity metric comprises maximizing a similarity metric.

12. The system of claim 10, wherein maximizing the proximity metric comprises minimizing a distance metric.

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13. The system of claim **9**, wherein mapping the query to the homogenized context vector, comprises using the large language model to classify the query based on a text label associated with the homogenized context vector.

14. The system of claim **9**, wherein the raw context vector is homogenized using local domains. 5

15. The system of claim **9**, wherein the raw context vector is homogenized using landmarks.

16. The system of claim **9**, wherein before using the augmented proximity metric, each vector embedding is associated with one of the homogenized context vectors of the plurality of homogenized vectors and stored in an extended vector database. 10

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